

# ripwire

## The ripgrep of AI context.

A zero-runtime-dependency C++23 CLI that maps any codebase into a ranked, deterministic call graph for coding agents — and puts a tripwire on every claim it emits.

### **rip** — the speed half

Structural answers in tens of milliseconds, from a call graph it built itself.

### **wire** — the honesty half

Unprovable totals ship labelled as floors. A zero means none found — never none exists.

// the problem

# Your agent reads the whole library to answer one question

Blind grep, then whole-file reads. Most of what enters the context window is never used — it is paid for anyway, on every step of every task.

And bigger context is not better context: irrelevant text actively degrades the answer. The job is selection — a small, ranked, structural map beats a pile of open files.

Speed caveat travels with the number: ripwire answers from a warm, pre-built index, and round 4 measured its index cost too — 0.31 s, tabulated beside the competitors' rather than omitted.

**0.108 s**

median answer, warm index — round 4, all arms one machine one day, N = 60

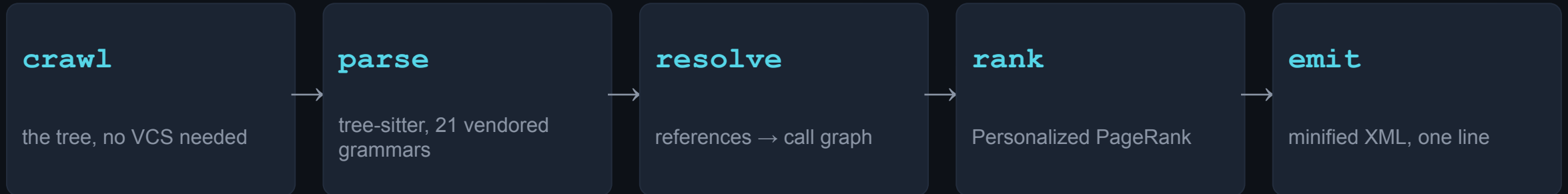
**−39.4%**

token ceiling vs the un-routed baseline — LocBench cost ledger, N = 243

// one binary, the whole pipeline

# Crawl → parse → resolve → rank → emit.

## Deterministic, end to end.



### byte-identical

two runs, same bytes — a gate on every push, not a tendency; warm equals cold

### zero runtime deps

CMake + a C++23 compiler; builds with the network off — vendored everything

### the languages

Rust · C++ · ObjC/C++ · C · Metal · CUDA · Python · Go · Swift · TypeScript · JavaScript · Java · Ruby · PHP · Lua · Bash · C# · JSON · TOML · YAML · Markdown — 21 vendored grammars; markdown headings are real symbols

### agent-native

an MCP server and 179 long flags behind one `--help` that is always the authority

// every feature, one screen

# Seven families of questions it answers

|                 |                                                               |                                                                                                                                                          |
|-----------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| understand cold | <i>What is this repo, and what matters in it?</i>             | <code>--for --tree --lego --exemplar --recall --pack-task --token-budget</code>                                                                          |
| navigate        | <i>Who calls this? Safe to change or delete? Which tests?</i> | <code>--callers --callees --uses --impact --path --connect --affected --situ --test-gate --from-trace --pattern --safe-delete</code>                     |
| detail ladder   | <i>Show me more — but only where it pays.</i>                 | <code>--detail --pack-signatures --outline --expand --compress</code>                                                                                    |
| quality & risk  | <i>Where is the risk, and did I just add some?</i>            | <code>--quality-panel --quality-delta --dmm --readability --ensemble --context-ratio --nonlocal-state --field-affinity --hotspots --lint --clones</code> |
| self-diagnosis  | <i>Is my setup actually working?</i>                          | <code>--doctor --skipped</code>                                                                                                                          |
| security        | <i>Is this agent skill file safe to install?</i>              | <code>--scan-skill --scan-skills</code>                                                                                                                  |
| knobs & modes   | <i>Shape, format, cache, budget.</i>                          | <code>--json --format --mcp</code>                                                                                                                       |

--help is generated from the binary's own flag table — 179 long flags; docs/COMMANDS.md carries an entry for every one of them, 161 with a recorded invocation and its output

```
// reach for it at the moment, not the manual
```

# The reflexes: which verb fires when

## Landing cold on a task

```
--for="<task>" · --pack-task
```

ranked, budgeted context in one call

## Stack trace in hand

```
--from-trace
```

frames mapped to symbols, innermost body included

## “Is it safe to change X?”

```
--impact · --uses
```

the whole blast radius, not just 1-hop callers

## Just edited a symbol

```
--edit-check
```

did the contract change, who breaks — ~ms, warm

## About to write a helper

```
--exemplar · --grep
```

imitate the house pattern; catch the duplicate early

## Landing several branches

```
--merge-scout
```

pairwise conflict sites + a landing order

## Before you call it done

```
--quality-delta · --test-gate
```

only what you made worse; exactly which tests to run

## Handing the area on

```
--handoff · --note-add
```

the verified-vs-heuristic packet the next agent needs

and `--pr-context` when you are reviewing someone else's diff — per-file blast radius, tests-to-run, hotspot flags

// new since 2026-08-15 — each figure re-derived on the binary that ships

# What 2026-08-15 to 2026-08-23 added

## **--pattern**

*search by code SHAPE, not by node kinds*

foo(\$X, ...) across 13 grammar objects / 11 languages. On this repo VERIFY(\$X) finds 101 hits over 625 eligible files, each with its enclosing symbol. A pattern no served grammar resolves REFUSES (exit 1) instead of reporting hits=0.

## **--safe-delete**

*“can I delete this?” in one call*

callers + transitive radius + every read/write/import site + how much of that radius a test reaches. risk= names what was FOUND — never a go/no-go verdict.

## **--impact importers=**

*the second, weaker reach*

the files that include/require a file defining SYM, with their own cap pair — NEVER summed into reaches=, because files and symbols are different units. IngestResult: reaches=0, importers=70.

## **compact --for bundles**

*bodies were 52.7% of every conceptual byte*

the subtoken+body route now ships the ranked map plus one-hop edge context, disclosed as bundle=compact bodies=0. 15-query class B: 184,857 B → 95,256 B, -48.5%, all 11 decisive markers still present. --auto-bodies is a permanent opt-out.

## **corroborated callers**

*shared= on --pack-task rows*

a neighbour reached by four of the bundle's anchors sorts above one reached by a single anchor — omitted at 1, which is what every 1-hop row satisfies anyway.

## **per-rule --lint roll-up**

*a capped view stops hiding whole rules*

on this repository 14 of the 31 rules that fired contribute ZERO shown rows — 368 findings whose only evidence is their count=. A rule no corpus language registers carries applicable=0, so its zero is inertness, not a measurement.

**And two languages: PHP and Lua**, each shipped with its floor stated rather than implied — PHP's run-time dispatch (\$fn(), call\_user\_func, \_\_call) names its callee at run time and is a declared floor; a Lua corpus reports no inheritance edges, because metatable inheritance has no syntax to read.

// new since 2026-08-23 — each figure measured, dated, and re-derivable

# What 2026-08-23 to 2026-08-30 added

## `--slice-flow`

*cross-statement data-flow slicing  
(ARISE, arXiv:2605.03117)*

The paper's own slicer — reaching-definition edges, a seed plus a direction, a bounded BFS that stops at function boundaries — implemented and MEASURED on a registered fix-commit protocol: flow rows lift function-level added-line recall 0.163 → 0.198 at 25% of --expand's whole-body bytes (whose recall is 1.0 by construction). 7 commits / 38 instances, cpp only — a thin corpus, reported as thin.

## `--slice, first numbers`

*the 2026-08-28 registered contract,  
finally measured*

Per-variable line-recall 0.726, hit-all rate 0.632 — a FLOOR under a noisy relevance oracle: every inspected miss was the protocol's word-regex matching identifiers inside comments and string literals, occurrences the slicer correctly refuses to call variable uses. No inspected instance showed a real occurrence the slice dropped.

## `~1350× on --expand`

*the secret-redaction sweep was  
quadratic in LINE length*

One selector in a 2.1 MB / 768-line minified yarn bundle NEVER completed — killed at 1,343.9 s of user CPU with an empty output file; 196.7 s is the only clean lower bound — and now answers in 0.46 s warm. The 20 KB single-line fixture: 23.02 s → 0.017 s. A pure memoization, so an output no-op: 120/120 invocations byte-identical; gated on behaviour, the timings a ledger row and never a red-CI threshold.

sources: docs/EVALS.md \$--slice-flow (registered protocol + per-instance ledger) · bench/PROFILE.md 2026-08-30 — no number travels without its caveat

// new since 2026-08-30 — the window this deck did not cover until now

# What 2026-08-30 to 2026-09-06 added

## **.gitignore, honoured by default**

*named for ripgrep, whose defining default this is*

The crawl walked ignored files; in a git work tree it now consults git's own rules and skips what the repository already declared uninteresting. This root, three checkouts under an ignored bench/external/: files= 8,674 → 1,522, cold 2.81 s → 0.52 s, warm 0.63 s → 0.10 s — and --no-ignore restores the previous walk exactly. Nothing drops silently: ignored\_files= is exact, ignored\_dirs= says the walk STOPPED there so those contents are UNKNOWN rather than zero, and both are ABSENT when the rules dropped nothing — so a tree with nothing ignored is byte-identical to what it produced before.

## **--html --color-by**

*the map, rendered —  
no server, no CDN*

One self-contained HTML file: a force-directed call graph you click to recentre. --color-by=lang|community|cx|churn|tested sets the INITIAL colour only — the page embeds all five and keeps a live selector, so switching lens costs no second run. One five-stop blue-to-orange scale rather than the usual green-to-red, so reading it never depends on telling red from green.

## **--eval-retrieval, re-sampled**

*the sampler was measuring  
the corpus, not the ranker*

For fourteen months the gold set was the first 150 doc-commented symbols in PATH order — and bench/ sorts before src/, so adding a docstring to a benchmark harness moved a published number without touching ranking code. One 60-symbol probe, one corpus, only its PATH varied: 0.105 and 0.356 MRR from spelling alone. Now exhaustive — population=3011 scored=3011 rule=exhaustive — and EVERY retrieval figure this project publishes was re-derived under it.

## **--plan-lanes**

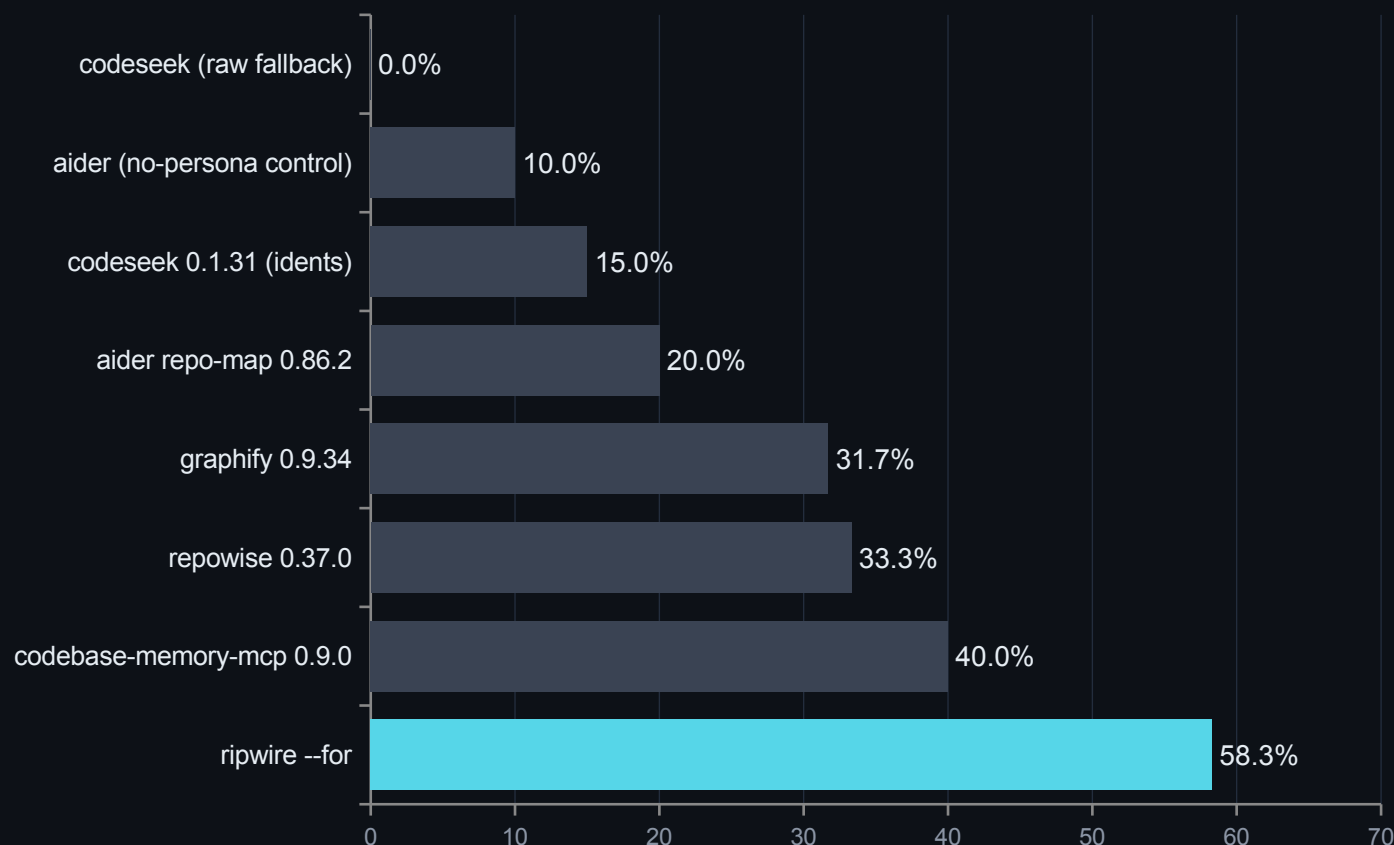
*which lanes would collide,  
BEFORE a line is written*

Where --merge-scout says "these branches already conflict", this says "these lanes WOULD conflict if assigned this way" — no ref to resolve, no re-ingest. Each lane now also carries an advisory execution profile (model + reasoning effort) that labels its own evidence basis="structural-only" and versions its policy separately from the schema, so a recommendation cannot quietly claim more than the structure it read.

CHANGELOG.md [Unreleased] carries the ignore-default measurement and its ledger in bench/PROFILE.md · the sampler negative is docs/EVALS.md §7, published as a counterexample against our own published numbers

// one harness, one machine, one day — N = 60 paired, zero exclusions

# Head-to-head: every competitor, one table



1.46x

the margin over the best competitor — 58.3% against 40.0%  
strict file@10

## And the index it was measured with

ripwire 0.31 s · codebase-memory-mcp 1.24 s · codeseek  
3.37 s · graphify 7.82 s · repowise 34.0 s, whose worst  
single index was **352 s**. Cold, nothing to an answer: 0.213 s.

**What re-running cost us, published:** aider moved 18.3%  
→ 20.0% on its own tie-break nondeterminism, and we  
printed the higher number. Paired, ripwire loses 2 instances  
to codebase-memory-mcp, 2 to repowise, and 1 each to  
graphify, aider and the aider control.

bench/headtohead/r4-2026-08-06/ — harness, per-instance JSONL and recipe committed · replaces the two non-comparable tables r1 and r2  
needed

// scored against a third-party oracle, never against each other

# Graded by a compiler: zero silent misses

Both tools were scored against a scip-clang compiler-grade index of this repository — 68 answers, 34 blind-authored queries × --uses and --callers.

0

true silent misses — pre-fix and post-fix alike. Every imperfect answer either carried a discriminating self-flag (amb=, defs>1, external="1") or was an oracle-scope artifact where ripwire was right and the compiler could not see the file.

## What a compiler-grade index costs

|             | ripwire | Serena 1.6.2.dev0 (clangd 19.1.2) |      |
|-------------|---------|-----------------------------------|------|
| warm query  | 49.8 ms | 3,760 ms                          | ~70× |
| cold start  | 208 ms  | 8.7 s                             | ~40× |
| index build | none    | ~8 min                            | —    |
| errors      | 0 / 101 | 7 / 61                            | —    |

**Read honestly in both directions.** The LSP is more precise — site-level 0.929/0.941 against ripwire's 0.914/0.941 after the fix round, a gap of roughly 1.5 points with recall now equal. ripwire also pays ~7× the bytes per call, because its output carries self-describing legends. What it cannot do is answer a macro query at all — clangd's documentSymbol has no macro entries, which is 3 of its 7 errors — or start in less than eight minutes.

**The number that moved the wrong way, published with the rest:** over-hedging rose 18.6% → 22.6% across the fix round. Closing loss buckets added self-flags faster than it removed imperfect answers. That direction is deliberate — calibrated to warn too often rather than too rarely — but it is a cost, and it is on the slide.

// 48 paired questions, zero exclusions, both arms exit 0 on all 48

# Against a graph-database context server: 27–7–14

| class                  | n  | ripwire | GitNexus | tie | bytes ÷ it |
|------------------------|----|---------|----------|-----|------------|
| symbol lookup          | 12 | 6       | 2        | 4   | 1.35×      |
| conceptual search      | 15 | 8       | 1        | 6   | 1.23×      |
| callers / blast radius | 12 | 7       | 2        | 3   | 0.39×      |
| task orientation       | 9  | 6       | 2        | 1   | 0.06×      |
| all 48                 | 48 | 27      | 7        | 14  | 0.159×     |

**The cost of the whole sweep**

**309,348 B** against 1,945,231 B. Warm median 197 ms against 1,082 ms. Index 0.25–0.45 s / 6.6–16.5 MB against 23–52 s / 391–623 MB — written into the repository, not \$TMPDIR.

**What it does better:** compact one-hop context at 0.7–1.0 KB, a depth-LABELLED blast radius, per-stage timing on every query, and two-command onboarding into eight agents.

**Its seven wins are named one by one, and its first-pass numbers do not exist.** Four are cost losses on an answer ripwire gets right — a by-name lookup it serves in ~1 KB where ripwire spends 3× to also hand back the body. Three are ranking misses on webpack, and two of those share one unfixed mechanism: a symbol whose NAME matches the query beats the implementation that carries it in its body. Under the improve-first rule the round's FIRST pass was never published — its product was a loss list, and what is on this slide is the state after that list was worked. One of those fixes met its pre-registered band and was reverted anyway for failing a separate standing requirement; **the loss it would have fixed is still counted against us above.**

**Failure modes, both directions**

|                            |   |        |
|----------------------------|---|--------|
| needed a 2nd call          | 0 | 8 / 48 |
| empty radius, real callers | 0 | 1      |
| malformed through a pipe   | 0 | 7 / 48 |

*The published numbers use the file-redirected form — the configuration favourable to it.*

// the rule: numbers are held back until the loss list is worked

# Every head-to-head round, and the ones that went against us

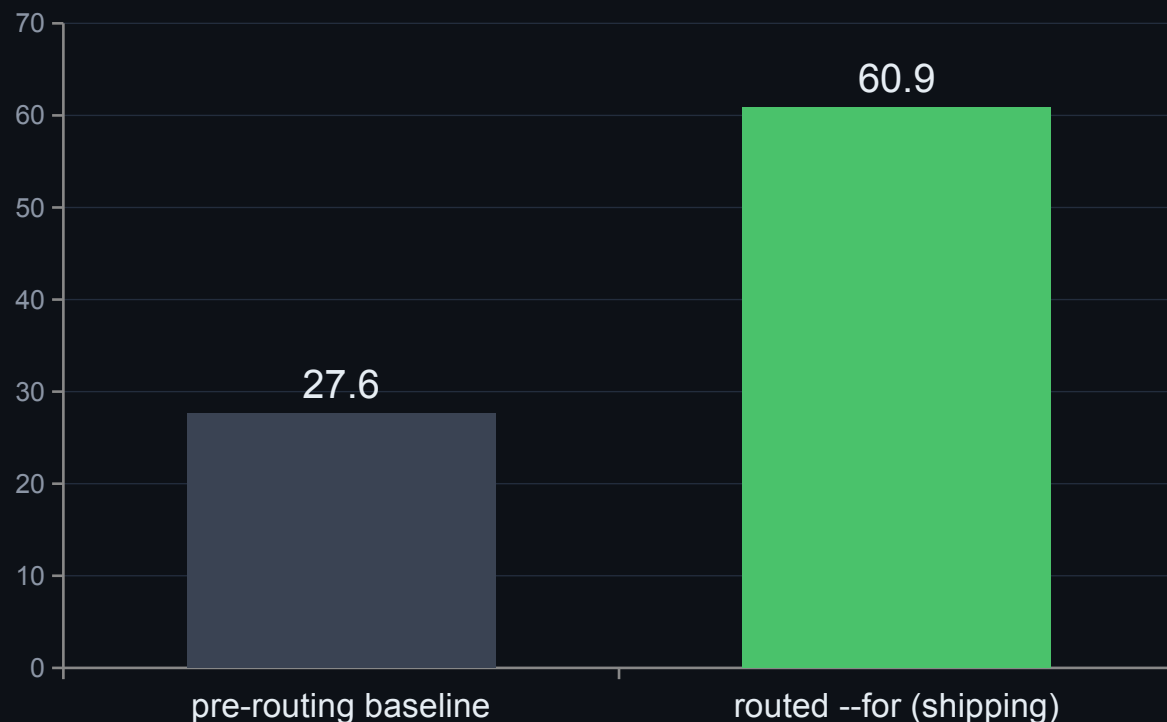
|     |               |                                                           |                                              |
|-----|---------------|-----------------------------------------------------------|----------------------------------------------|
| r1  | 2026-07-13/14 | Aider repo-map · codebase-memory-mcp · graphify           | superseded by r4                             |
| r2  | 2026-08-03    | repowise · codeseek                                       | superseded by r4                             |
| r3  | 2026-08-03    | headroom — a compression layer, so a different instrument | passed code through byte-identical           |
| r4  | 2026-08-06/08 | all five competitors, one harness, one day                | the table on the previous slide              |
| r6  | 2026-08-04/05 | agent-in-the-loop: the Codex CLI pilot                    | localization parity; +80% tokens, classified |
| r7  | 2026-08-08    | CodeGraph, loss-first — we looked for our losses          | fix REJECTED by its own preregistered band   |
| r8  | 2026-08-08    | aider again, at equal token budget                        | loss questions traced to one root cause      |
| r9  | 2026-08-09    | Serena / scip-clang oracle                                | 0 silent misses; the fix list it produced    |
| r10 | 2026-08-22    | GitNexus 1.6.9 — a graph-database context server          | 27-7-14 — its own slide, two back            |

**r7 in full, because it is the point:** the fix was built, gated green, measured — predicted 12/22 against a pre-registered band of 10–14, measured 5/22 — and reverted rather than tuned. Published as a negative in docs/EVALS.md §7.

bench/headtohead/ (r1-r4, r9) · bench/r7/ · bench/r8/ — harnesses committed even for rounds that rejected their own fix · r5 left no head-to-head record in this tree, so it is not listed · r10's method and pins are published in docs/EVALS.md §2, its pre-fix numbers nowhere

// held-out, repo-disjoint, frozen dataset

# Routing the ranker: +33 points, measured properly



**+33.33pp**

paired delta, 243 held-out instances in 78 repository clusters

**+25.00pp**

clustered-bootstrap 95% lower bound — the honest floor

**-39.4%**

token ceiling for that gain

**+3.4%**

warm latency for that gain

// compiled, not interpreted — and measured on public corpora

# Fast enough to call on reflex

**0.213 s**

cold — nothing to an answer: parse, rank,  
reply, no cache (r4)

**0.108 s**

warm query, median — against aider's 2.920  
s (r4)

**0.31 s**

index build — against repowise's 34.0 s,  
worst case 352 s (r4)

**49.8 ms**

warm query on this repo — against a clangd  
LSP's 3,760 ms (r9)

## Where the second goes

Cold profiles are tree-sitter parse plus tag-query capture — not graph math. Resolve and PageRank together are a small fraction of the run, and the ranking was never the bottleneck.

Which is why the wins came from removing work, not micro-optimising it: demand-driven side captures, five whole-AST passes fused into one pre-order walk, a calibrated per-worker parse reserve.

## Two opt-in faster builds

LTO is on by default. PGO is a script away and buys 14–25% cold.

Both are opt-in performance, not opt-in correctness: **the output stays byte-identical across build flavours** — the determinism gate runs on both, and CI builds a plain flavour too, because NDEBUD once compiled a whole class of degrade-path checks out of existence.

every card above comes from a committed harness — bench/headtohead/r4-2026-08-06/ and r9-2026-08-09/ — not from a private corpus

// how it got there — the pre-release performance sprint

# Work removed before work optimized

## file ingest

FILE\* whole-file reads; CSV counting became a byte scan — iostream out of the hot path

## capture gating

value-use capture runs only for the verbs that actually need it, not on every run

## AST pass fusion

five whole-AST side-capture passes share ONE pre-order walk instead of five

## parallel parse

query compile overlaps parse scheduling; files schedule by size — ids stay deterministic

## radix + S+tree

numeric keys for edges and scores; a bounded-scratch hot map whose fixed cap reports its own overflow

## no std::map

unordered\_dense, reserve() before ingest — no std::unordered\_map on any hot path

**Why this slide carries no numbers.** The sprint's own headline pair was measured in June 2026 on a large private C++ corpus that is not publicly reproducible, and the deck does not quote figures a reader cannot re-derive — the previous slide's numbers all come from committed harnesses. What survives the omission is the **shape**: the graph math was never the bottleneck, and every win came from doing less work rather than doing the same work faster. Same map contract throughout — byte-identical output before and after each of these changes.

the mechanisms are in bench/PROFILE.md and the commit history; the private-corpus figures they produced are deliberately not quoted

// docs/EVALS.md \$5, pinned on this repository 2026-08-08 (the --for row re-pinned 2026-08-22)

# Ten everyday moments, and what each one costs

| Ask it                                 | Command                   | ripwire    | naive read | savings      |
|----------------------------------------|---------------------------|------------|------------|--------------|
| Orient me in this repo                 | ripwire .                 | ~5.6K      | ~20K-25K   | 3.6x-4.5x    |
| Where is X handled?                    | --for="..."               | ~2.1K      | ~4.9K-20K  | 2.3x-9.3x    |
| What do I already know?                | --recall="..."            | ~15K       | ~445K      | 29.2x        |
| Set me up for this task                | --pack-task="..."         | ~2.1K      | ~16K-80K   | 7.7x-37.7x   |
| Show me this one function              | --expand=SYM --top-k=0    | ~260-16.5K | ~43K-174K  | 2.6x-670x    |
| Who calls this function?               | --callers=SYM             | ~580       | ~40K-52K   | 69.2x-89.1x  |
| Is it safe to change this?             | --impact=SYM + --uses=SYM | ~1.3K      | ~18K       | 14.4x        |
| I have a stack trace                   | --from-trace=FILE         | ~1.4K      | ~124K-298K | 86.9x-208.6x |
| I changed these files — tests? radius? | --situ                    | ~410       | ~3K-132K   | 7.3x-324.2x  |
| Review this PR/diff                    | --pr-context=REF          | ~1.9K      | ~4.8K-51K  | 2.6x-27.5x   |

Figures are ~tokens (≈ bytes/4). Every row is scored same-correct-answer-or-it-doesn't-count — both sides were checked, not assumed — and every ratio is a **range**: the cheap end and the honest end of what an agent would actually read, never the single most flattering number. Absolute counts belong to the corpus AT THE PIN — .gitignore became a crawl default on 2026-09-03, which moved what a re-run reads; the commands, not the constants, are what reproduces.

// the cost lever

# Smaller answers that are also better answers

## 74.7%

fewer element bytes at top-50 — the --pack-signatures ladder,  
root-neutralised, re-derived on this repo every run

Ask for a symbol by NAME and the router uses the  
name-exact ranker:

**MRR** 0.745 → 0.960

**recall@1** 61.1% → 91.3%

name-shaped queries on src/, all 3,011  
doc-commented symbols scored, re-pinned  
2026-09-05

This figure survived its own audit: three corrections deep, re-derived  
root-neutrally after the original was shown to depend on how the  
corpus path was spelled — three spellings of one root read 18.6  
points apart before that subtraction. top-50 is the quotable number  
because the payload is top-50 whatever --top-k says.

And the pollution metric — fixture and generated paths  
contaminating results — is driven to 0.0% on the ranking lane  
while recall went UP, not down.

docs/EVALS.md \$4-\$5 — showcasecapturecheck re-derives the byte-reduction triple; `ripwire src --eval-retrieval` re-derives the ranking  
bars, scoring every doc-commented symbol rather than a sample

```
// the half no other tool ships
```

# The tripwire: honesty as an output contract

## `counts_floor="1"`

call edges come from source text — dynamic dispatch adds no edge, so every count that cannot claim completeness is labelled a floor, in the output itself

## `a zero is a measurement`

zero means none found, never none exists — and absent  $\neq$  0

## `truncation is disclosed`

every cap, page seam and budget cut is named in the header before you read a row

## `refusals teach`

a refused query names the flag, the problem, and a working example — never a bare error

```
$ ripwire . --callers=rankGraphTeleport

<callers of="rankGraphTeleport"
  defs="1" count="6"
  counts_floor="1">
<s t="fn" n="runEval" .../>
<s t="fn" n="rankGraph" .../>
...
</callers>
```

**The map grades itself before it answers.** This repository's own src/: files=153 symbols=5122 edges=14182 ambiguous=5982 unresolved=1598.

```
// the honesty marks stop being a promise and become a measurement
```

# Four levers accepted, one rejected on purpose

## macro edges

kParserVer 48

function-like #define invocations land role="macro" edges on disclosed-degraded symbols

## fn-pointer bindings

kParserVer 49

calls through unambiguous local function-pointer bindings now resolve

## using-declarations

kParserVer 51

re-exports emit role="import" references — r9 loss bucket 1

## shadow suppression

kParserVer 52

a local variable shadowing a function name stops stealing its use-sites — r9 loss bucket 2

## RTA-lite

reverted

instantiation-filtered CHA cone: refuted TWICE — the instantiated set crossed namespaces, and narrowed calls LOST their amb= mark

The two r9 levers, against the compiler oracle: **site precision 0.9046 → 0.9136, recall 0.9285 → 0.9412.**

**The fn-pointer lever raised unresolved= by 1,039. Disclosure, not regression:** call sites the resolver now RECOGNIZES but refuses to resolve are counted instead of silently ignored.

// how it stays true

# Proven, not promised

## 547 gate scripts

the suite runs on every push — plus determinism, cache-transparency and golden contracts; the gate count itself is gated against the runner's own loop

## byte-identical, always

two runs over the same tree produce the same bytes; warm equals cold. Enforced in CI, twice — Release AND a plain flavour, because NDEBUB once blinded a whole class of checks

## differential refactoring

a refactor must prove it changed nothing observable: two binaries, hundreds of argv vectors, stdout + stderr + exit codes byte-identical

## held-out labels, authored blind

eval labels were written by reading source before the ranker ever ran on them — so the eval is allowed to say the ranker is wrong. It has.

## sanitizer wall

ASan + UBSan + integer + float-cast, no-recover; TSan separately; leak suppression is one pinned file

## its own output is audited

the showcase capture is regenerated and adversarially re-read as a claims corpus — the methodology ships in docs/METHODOLOGY.md

// the reason to believe any number on this deck

# Claims you can trust, because we publish what failed

547

gate scripts named by test/regression.sh — and the COUNT itself is gated against the runner's own loop, so it cannot go stale quietly

8

registered NEGATIVES — changes built, gated green, measured against a band written before the code, and reverted rather than tuned

0

numbers on this deck without a committed instrument behind them — §8 lists the ones this project refuses to publish at all

## the fresh-clone arm

our own battery only ever ran inside a configured, long-lived worktree — one value of every AMBIENT input. It clones HEAD into a genuinely unconfigured checkout and re-runs the gates whose past failures were exactly that: a developer's git config, a fixture's inherited branch name, the checkout's own path length.

## the merge-conservation gate

consistency is not conservation. A conflict resolution can drop a gate from the runner's loop while regenerating the pinned count from that SAME damaged list — both sides then agree at a wrong number. On a merge, the loop must equal the UNION of both parents', minus explicit tombstones.

## the eight, by name

r7's prose-strip fix (predicted 12/22, band 10–14, measured 5/22) · RTA-lite, refuted twice · file-level evidence pooling, +0.00pp held-out · un-guarded query stemming · its IDF-guarded retry, VOIDED by its own verifier · query-term density weighting, both displacement guards tripped · --anchor and --cochange-boost, no confirmed lift · a name-coverage floor that MET its pre-registered band and was reverted anyway for a standing requirement.

*A gate that cannot go red is a gate that proves nothing — which is why every arm on this deck ships with the mutation that makes it fail.*

// own the losses — they are in the README too

# Where it loses, published with the wins

**The public C++ number is lower than the private one it replaced**

SFML: strict file@10 28.7%, any@10 41.7%, first-hit MRR 0.21. Until 2026-08-07 this bullet compared against a ~89% any@10 private corpus that is no longer reproducible. That comparison is retired; the public number is the baseline going forward.

**--grep costs more than it saves**

+19.7% tokens / -11.2% — published next to the flag it indicts, and carrying the same private-corpus caveat as every figure in that source

**PageRank is the wrong co-change ranker**

3.8% recall@5 against 40.3% for plain lexical, and fusing the two made it worse — relatedness is lexical, importance is structural

**Strict multi-file localization stays hard**

single-file gold 73.4% vs multi-file 18.2% (held-out); the same cliff on every corpus — open headroom, said plainly

*A tool that publishes its counterexamples is a tool whose wins you can believe.*

```
// the single command for "is this code actually rotten?"
```

# Six families of evidence, counted — never blended

**structural**

shape: complexity, size, nesting, params, readability rank

--metrics

**lexical**

identifier text: the naming rules

--lint --naming-consistency

**confusion**

syntax: the atoms-of-confusion rules

--lint

**historical**

git change frequency, measured PER FILE

--hotspots

**colocation**

how much you must READ outside this file

--context-ratio

**state**

this function's OWN BODY touching non-local mutable state

--nonlocal-state

**Ranked by the COUNT of distinct families that fire** — never a weighted composite, because a composite lets one loud family impersonate six.

**4 of 6** is what the strict preset counts — the verb says so itself, families="6" enabled\_n="4". Two families did not clear the stability ladder.

--quality-panel[=strict|default|lenient] — and --quality-delta for the narrower question: what did MY change make worse?

// a new lens does not ship because it sounds plausible

# The calibration that disqualified a family

**+0.168**

the LARGEST cross-family correlation in the whole 6×6 matrix — widening the panel from four families to six did not raise it

**27,889**

eligible functions, five independent trees — two reported separately and never pooled, because one of them is this repository

**4 of 6**

families the strict preset actually counts — TWO were measured and held back from gating

**The uncomfortable result, kept.** The stability pass ran each family across a ladder of commits to ask whether it is steady enough to gate on. **colocation came out worse than historical on one ladder**, so neither gates: strict counts four of the six — a lens its own author wanted, measured, and demoted. The ladder was run precisely because that answer was possible; assuming it would pass is how a plausible axis becomes a permanent one.

**--field-affinity was NOT made a family either** — an exclusion by unit, not a failed measurement: it ranks struct FIELDS by co-access against declared layout, and the panel's unit is a function. Stated in the calibration rather than left for someone to notice.

```
// built for the agent's seat
```

# One command wires it into your agent

```
$ ripwire wrap claude
```

claude · cursor · codex · windsurf · gemini · aider — or --all to detect every one you have installed

## 31 MCP verbs

16 read verbs mirroring the CLI, 12 flagship reflexes (impact, uses, edit\_check, from\_trace, connect ...), 3 span-addressed edit verbs with a safety contract

## lazy-body handles

read verbs return signatures and a stable handle; the agent fetches a body only when it decides it needs one — names by default, bytes on request

## 18 agent skills

moment-matched workflows (orient, navigate, change-check, quality-bar ...) — wrap prints the recipe, skills/install.sh installs them

## 11 orchestrator loops

copy-paste prompts in prompts/: run the same audit, eval and head-to-head machinery that built this tool, on your own repository

**wrap does not just print JSON:** it probes what that agent already has, emits the config in that agent's own shape, and includes a use-when blurb so the agent knows which verb fires at which moment — not just that a server exists.

the MCP server exposes the same deterministic engine — one index, shared with the CLI, staleness-checked

```
// it does not only read
```

# Span-addressed edits: the map writes back

## Three verbs, one resolved span

```
--replace-symbol-body=TARGET  
--insert-before-symbol=TARGET  
--insert-after-symbol=TARGET  
--edit-payload=FILE|-
```

TARGET is a name, an @FILE:LINE seed — paste the location out of a diff hunk or a compiler error and it binds the innermost enclosing definition — or a freshness-pinned handle from --grep --handles. No line numbers to keep in your head, and no whole-file rewrite.

## The refusals are the feature

Freshness hash, lock, a re-check immediately before the rename, fsync, mode preserved, atomic rename. A target that resolves to more than one definition refuses. An empty payload refuses rather than being read as a deletion. Every refusal leaves the file byte-identical, and every success prints a JSON receipt whose span is the POST-EDIT byte range — where the payload now sits, not where it was aimed.

## Preview the bytes before they exist

--edit-check=SYM --edit-payload=- --dry-run splices the payload over the span IN MEMORY, re-parses that one file, rebuilds the call graph over the re-derived tree, and answers the contract question about bytes nobody has written: did the signature change, and which callers does it break? Preview, then apply — no Read of the file in between.

## Several edits, one transaction

--edit-plan=FILE is a versioned JSON multi-edit plan and its mode is EXPLICIT: exactly one of --dry-run or --apply, and neither is not a default. Every target, payload and span is preflighted before any write; overlapping edits refuse; payload paths are confined to the plan's own directory, so an absolute path or a '..' escape refuses and names what it resolved to.

**Same engine on both seats:** the three MCP edit verbs are these three flags, and the navigation the deck has no room for is the same story — --graph-query, --whereis, --stray-content, --at, --arch, --owners, --export, --help-task all sit in docs/COMMANDS.md with a recorded invocation and its output.

every claim on this slide is stated by `ripwire --help` and captured in docs/COMMANDS.md — test/deckcheck.sh proves each flag named here exists in the shipped binary

// standing on giants

# The research inside — classic and current

41 repositories + 67 papers folded · and a labelled survey of 237 tools that contributed nothing, which says so — every row with the lesson taken and where it lives: docs/LINEAGE.md

|      |                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------|
| 1976 | <b>Cyclomatic complexity — McCabe</b><br>the metric behind --hotspots                                 |
| 1994 | <b>Okapi BM25 — Robertson · Spärck Jones</b><br>the lexical ranker (twice: subtoken+body, whole-name) |
| 1998 | <b>Personalized PageRank — Page · Brin</b><br>the headline importance signal                          |
| 2008 | <b>Louvain modularity — Blondel et al.</b><br>the module / community map                              |
| 2009 | <b>Reciprocal Rank Fusion — Cormack et al.</b><br>deterministic signal fusion (--rank-by=rfr)         |
| 2011 | <b>Readability model — Posnett · Hindle · Devanbu</b><br>the lexical family of the quality panel      |
| 2019 | <b>Delta Maintainability Model — di Biase et al.</b><br>--dmm: one comparable number per change       |

counts gated in-repo (readmedriftcheck arm E derives them from LINEAGE.md's own tables) — the lineage is the design rationale, not decoration

|                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TDAD · arXiv 2603.17973</b><br><br>a static test-map cut agent-caused regressions 6.08% → 1.82%; prose TDD instructions alone made agents WORSE → the shape of --test-gate |
| <b>What-to-Retrieve · arXiv 2503.20589</b><br><br>retrieval selection beats retrieval volume for coding agents → the selection-over-dumping thesis                            |
| <b>LocAgent / Loc-Bench (2025)</b><br><br>the localization metric (strict Acc@k) and the frozen 560-instance dataset every accuracy number here is scored on                  |
| <b>scip-clang · Serena / clangd</b><br><br>the compiler-grade oracle round 9 graded both tools against — an outside referee, not a rival                                      |
| <b>tree-sitter</b><br><br>the incremental GLR parsing substrate — 21 grammars vendored, one parser per thread                                                                 |

// do not take any of it on trust

# Every claim, and the command that re-derives it

179 long flags · 29 slides

```
bash test/deckclaimcheck.sh
```

every --flag named here exists

```
bash test/deckcheck.sh
```

74.7% fewer element bytes

```
bash test/showcasecapturecheck.sh
```

547 gate scripts

```
bash test/manifestcheck.sh
```

41 repos · 67 papers · 237 surveyed

```
bash test/readmedriftcheck.sh
```

the ten moments, any row

```
ripwire . --callers=SYM | wc -c
```

the head-to-head table

```
bench/headtohead/r4-2026-08-06/
```

the oracle round

```
bench/headtohead/r9-2026-08-09/RESULTS.md
```

**The deck is generated, and the generator is gated.** `present/deck5_ripwire_build.js` is scanned for fabricated flags and its slide count is derived from its own `addSlide()` calls — a deck that drifts from the binary fails the suite.

`docs/EVALS.md` is the source of truth; `$7` is the counterexamples, `$8` the claims this project refuses to make

```
// sixty seconds to the first ranked map
```

# Quickstart

```
git clone <repo> && cd ripwire  
cmake -S . -B build && cmake --build build -j          # hermetic: works with the network off  
./build/ripwire --help
```

```
ripwire .
```

the ranked map — start here on any unfamiliar repo

```
ripwire . --for="cache invalidation"
```

the task lens: what to touch, ranked

```
ripwire . --callers=someFunction
```

who calls it — with the honesty marker attached

```
ripwire . --quality-panel
```

is this code actually rotten? six families, one shortlist

```
ripwire . --test-gate
```

before you commit: exactly which tests must run

```
ripwire wrap claude
```

wire it into your agent, in that agent's own config shape

**ripgrep** for the speed. **tripwire** for the honesty. Apache-2.0.